TTTC2343 Project Instruction: Customized Robot Simulation Development

Develop a customized TIAGo robot simulation using ROS and Gazebo. The project includes environment development, autonomous navigation, and robot task execution such as QR code scanning or pick-and-place operations.

Week 7: Draft Proposal

- Requirements:
  - Project title
  - Problem statement
  - Objectives
  - Proposed environment
  - Robot task
  - System flowchart
- Deliverables:
  - Proposal document
  - System architecture

Week 8: Customize Environment in Gazebo

- Requirements:
  - Minimum 3 areas/rooms
  - Obstacles and task stations
  - TIAGo robot integration
- Deliverables:
  - Gazebo world file
  - Screenshots
  - Launch file

Week 9: Mapping and Multi-Point Navigation and UI Development

- Requirements:
  - Generate map
  - Minimum 3 navigation checkpoints
  - Obstacle avoidance
  - User interface (UI) for robot control
    - Receive user input for destination selection

    - Navigate autonomously to selected location
    - Display robot status through UI

- Deliverables:
  - Map files
  - Navigation demo video
  - ROS navigation package

Week 10: Perform Task Action

- Requirements:
  - QR code scanning OR Pick-and-place operation
- Deliverables:
  - Functional ROS package
  - Task demo video

Week 11: Finalize Project and GitHub Upload

- Requirements:
  - Integrate all modules
  - Record full simulation
  - Upload project to GitHub
- Deliverables:
  - GitHub repository link
  - Final simulation video

Final Demo, Report, and Presentation

- Final Report:
  - Introduction
  - Methodology
  - System Design
  - Navigation and Task Execution
  - Results and Discussion
  - Conclusion
- Presentation:
  - 10–15 minute presentation
  - Demo and Q&A session

Suggested Workflow

Gazebo Environment
↓
SLAM Mapping
↓
Autonomous Navigation
↓
Perform Robot Task, UI Development
↓
Simulation Demo
↓
GitHub Submission

Evaluation Scheme: Customized TIAGo Simulation Project

Overall Marking Scheme

Component	Marks
Proposal	10
Gazebo Environment Development	15
Mapping, Navigation, Task Execution and UI Development	35
System Integration, Demo & GitHub	15
Final Report	15
Presentation & Demonstration	10
Total	**100**

Final Report Marking (15 Marks)

Criteria	Marks
Technical Content & Methodology	5
System Design & Implementation	5
Results, Discussion & Report Quality	5

Presentation Marking (10 Marks)

Criteria	Marks
Technical Explanation	4
Simulation Demonstration	4
Presentation Delivery & Q&A	2